Know How

Q&A Section

Q I am new to the world of Linux. I have used it on my desktop and am comfortable doing things with the graphical interface. However, a few days back I needed to rename a few files and folders on a remote computer from the command line. Please advice me on how to go about this.

—Sneha, Kolkata

To rename files and folder in Linux, you can use the *mv* command. For example, to rename a file called *test.txt* to *testnew.txt*, you can run the following command:

```
mv test.sh testnew.sh
```

Similarly, you can do this for folder/directory names too. You can even use wild cards with commands as well, if you need to rename multiple files/directories.

Q Recently, in an interview, I was asked if there was any tool to get the names and list of packages installed on Debian, Red Hat, Mandrake and SUSE. Is there any such specific tool that is used for all these distros, or are they different?

—Deeksha Chaudhary, by e-mail

You can use the command *dpkg -l* for Debian and Ubuntu. For Red Hat, Fedora, Mandriva and SUSE, you can use *rmp -qa*.

Q I was trying to schedule some scripts to run automatically. Can you please help me understand the different settings that need to be done to set up a scheduler in Linux.

—K. Singh, Patiala

Scheduling a task/job is done using a utility called *cron*, which makes tasks automatically run in the background at regular intervals. To manage *cron* jobs, there is *crontab*, a file that contains the schedule of *cron* entries to be run at specified times. To set up a scheduler, you need to make entries in this. Executing *crontab -e* opens the file in editable mode so that you can enter the details and save it. The *crontab* syntax has five fields as mentioned below.

```
*   *   *   *   *   command
```

Here, you can replace the first asterisk to enter the day of week—0 to 6, where 0 is Sunday. You can enter the month (1-12) in place of the second asterisk. Replace the third asterisk with the day of month (1-31). The fourth one is for the hour (0-23), while the fifth one is to enter the minutes (0-59).

For example, if you need to run a script daily at 5:30 p m, then the entry will as shown below:

```
30   17   *   *   *   sh /home/myuser/
scripttorun.sh  >/dev/null 2>&1
```

If the last part ">/dev/null 2>&1" is omitted, then by default, *cron* will send an e-mail to the user account after executing the *cronjob*.

Q I have an old Fedora installation on my system. I have just upgraded my RAM from 256 to 512 MB. My swap partition is also of 512 MB. Is there any way by which I can increase my swap space to 1 GB without formatting and reinstalling my OS.

—Amit Nandam, Gaya

Sure you can increase the size of swap without reinstalling the OS. This can be done by either adding a new swap partition or creating a swap file, instead. Here, I will discuss the steps to add a new swap file to increase the swap size as this is possible even if you do not have free unpartitioned space in your hard disk for creating new partitions, but have free space on one of your partitions.

To make a swap file of 1 GB, multiply 1024 MB and 1024 to get a block size. Now, at a shell prompt, as the root user, type the following command with the count being equal to the desired block size:

```
dd if=/dev/zero of=/swapfile bs=1024
count=1048576
```

Now set up the swap file:

```
mkswap /swapfile
```

Enable the swap file after creating it by using the following command:

```
swapon /swapfile
```

Add the following entry in the the */etc/fstab* file to make the system activate this file as swap while booting the system:

```
/swapfile  swap    swap    defaults      0 0
```

This will enable your 1 GB swap space, every time your system boots. You can check the size of swap by using the *free* command or by *cat /proc/swap*. END